can you give me steps to do this

Here's a **step-by-step plan you can follow in 2–3 hours** to build a working **Exam-Mode Privacy Coach** MVP using Cactus on-device AI. [1] [2]

1. Goal for the hackathon (MVP)

By the end, you should have:

- A mobile app with one screen.
- Button: *Record answer* → records audio → shows transcript + on-device AI feedback.
- Everything runs **locally** (no cloud, no login).

2. Step 1: Choose your stack (5 minutes)

Pick one you're fastest with and stick to it:

- **React Native** (recommended if cross-platform).
- **Flutter**.
- **Android (Kotlin)**.
- Or **simple web app** if you want to ship absolutely anything fast.

3. Step 2: Set up the project (10–15 minutes)

- **Create a new app**
    - React Native: `npx react-native init ExamPrivacyCoach`
    - Flutter: `flutter create exam_privacy_coach`
    - Android: Create a new project in Android Studio.
- **Add a simple screen**
    - One screen with:
        - A big button: **"Record Answer"**.
        - A text area: **"Transcript: ..."**.
        - A feedback box: **"Feedback: ..."**.

4. Step 3: Integrate Cactus SDK (20–30 minutes)

- **Install Cactus SDK**
  - Follow the quickstart from: https://www.cactuscompute.com (SDK install instructions for your platform).[2]
- **Load models on-device**
  - Pick:
    - An on-device **speech-to-text** model.[1] [2]
    - A small **on-device LLM** (3–4B class) via Cactus.[3] [1]
- **Test that the model can run locally**
  - Run a simple “Hello” test (e.g., pass a short text to the LLM and read the output).

5. Step 4: Add audio recording (15–20 minutes)

Depending on your stack:

- **React Native**
  - Use a library like `react-native-sound` or `expo-av` for recording sound.
  - On button press: start recording → stop on second press → save audio file.
- **Flutter**
  - Use `record` or `just_audio` to record audio.
- **Android**
  - Use `MediaRecorder` (minimal UI: Start / Stop).

Goals:

- When user taps **“Record Answer”**, start recording.
- When user taps again or after 60 seconds, stop recording and save the file path.

6. Step 5: On-device speech-to-text (15–20 minutes)

- **Pass audio file to Cactus STT**
  - Use Cactus’ audio-in → text-out API (no cloud upload).
  - Example pattern:

    ```
    const transcript = await cactus.stt.fromFile(audioPath);
    ```

- **Display the transcript**
  - Put the transcript text in your “Transcript” box on screen.

7. Step 6: Call on-device LLM for feedback (20–25 minutes)

- **Design the prompt**
  Example prompt you send to the on-device LLM:

  ```
  You are an exam-mode coach for students. Your job is to give short, constructive feed

  User's answer:
  "{{transcript}}"

  Task:
  - Score this answer from 1–10 on clarity, structure, and completeness.
  - List 2–3 short bullet-point suggestions.

  Output format:
  Score: 7/10
  Feedback:
  - Be more concise in the introduction.
  - Add at least one concrete example.
  ```

- **Call the LLM**
  - Use Cactus' on-device LLM API:

    ```
    const feedback = await cactus.llm.generate(prompt);
    ```

- **Parse the response**
  - Split by line or regex to get:
    - `Score: X/10`
    - Each bullet under `Feedback:`
  - Store this in state and update the "Feedback" box.

8. Step 7: Build the UI flow (15–20 minutes)

Make sure the user flow is smooth:

- **Initial screen**
  - Title: "Exam-Mode Privacy Coach"
  - Subtitle: "Everything stays on your phone, no cloud, no tracking."
  - Button: "Record Answer"
- **After recording**
  - Show loading indicator while STT + LLM run.
  - Then show:
    - Transcript
    - Score (e.g., "Clarity: 7/10")
    - Bullet-point feedback

- **Optional “Try again” flow**
  - Add a “Try again” button that resets the values and lets the user record another answer.

9. Step 8: Add privacy + “on-device only” branding (10 minutes)

- **On-screen copy**
  - Add text like:
    - “All audio and text stay on your device.”
    - “No data sent to the cloud.”
- **No login, no account, no backend**
  - Make sure your app has **zero** HTTP calls to external servers.

10. Step 9: Polish for demo (10–15 minutes)

Before recording your demo:

- **Test with a real example**
  - Record yourself: “Tell me about a time you solved a hard problem.”
  - Confirm:
    - Transcript is readable.
    - LLM gives a short score + 2–3 bullets.
- **Add a name and simple logo**
  - e.g., “ExamGuard” or “PrivacyCoach”.
- **Record a 1–2 minute walkthrough**
  - Steps:
    1. Open app → read privacy line.
    2. Click “Record Answer”.
    3. Speak a short answer.
    4. Show transcript + score + feedback.
    5. Say: “All of this happens offline, on the device.”

11. Time-boxed plan (roughly)

Time (approx)	What to do
0–15 min	Pick stack + create app + basic UI
15–30 min	Install Cactus SDK + test on-device LLM
30–45 min	Add audio recording

Time (approx)	What to do
45–60 min	Integrate on-device STT → show transcript
60–80 min	Add LLM prompt + parse feedback
80–90 min	Polish UI + privacy copy
90–120 min	Record your demo + tweak flow

If you tell me your exact stack (e.g., “React Native on iOS”, “Android Kotlin”, or “web app”), the next step can be **exact pseudo-code** for each of those pieces so you can copy-paste and adapt quickly.

⁂

1. https://securitybrief.co.uk/story/global-hackathon-champions-private-on-device-personal-ai-apps
2. https://www.cactuscompute.com
3. https://cerebralvalley.ai/e/executorch-hackathon